

Statičko izvršavanje

Determinizam

Izračunavanje

- Povećanje gustine izračunavanja za postojeće energetske mogućnosti
 - Wage-Scale – Cerebras
 - IPU – GraphCore
 - TSP – Groq
 - Wormhole - Tenstorrent

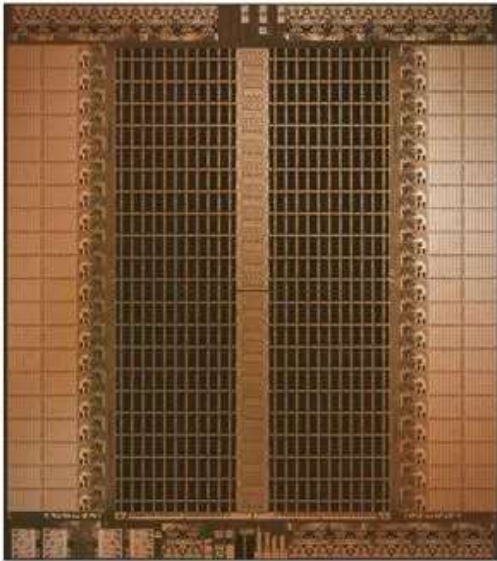
Statično ili Dinamično

- Statičko izvršavanje
 - Računanje sa gustim matricama
 - Većina AI
 - Računanje u finansijama
 - Industrijska/sensorska obrada
 - Kontrola u realnom vremenu
- Dinamičko izvršavanje
 - Retke matrice
 - Grafovski algoritmi
 - Veb server
 - Igrice
 - Procesiranje transakcija

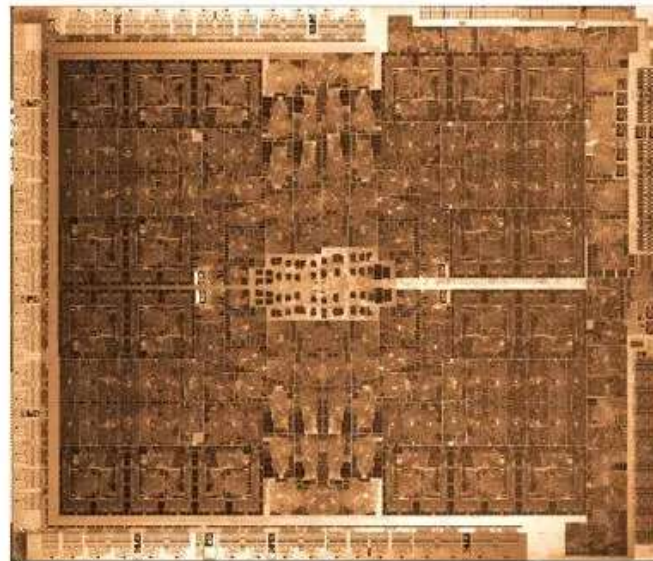


Statički ili dinamički procesor (1)

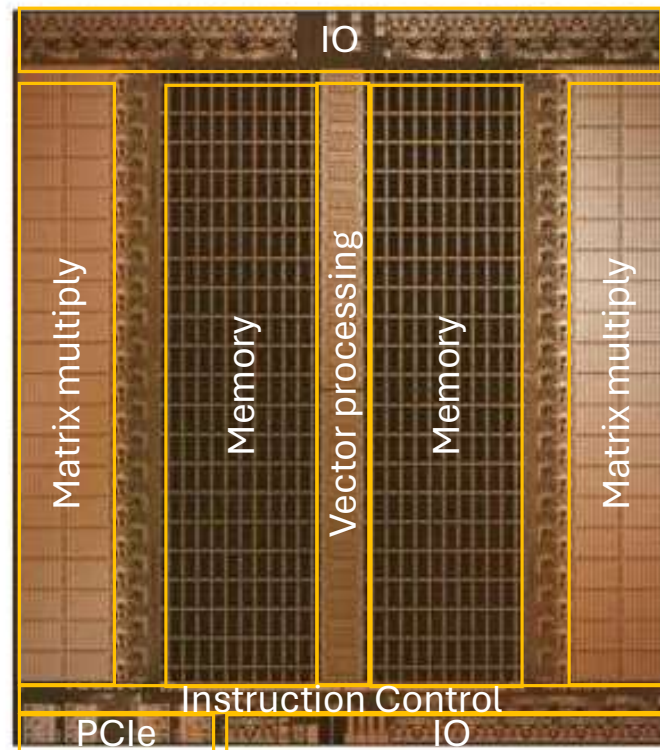
———— GroqChip™ 1 ————



———— Typical GPU ————



TSP arhitektura



Detalji arhitecture

SRAM Memory

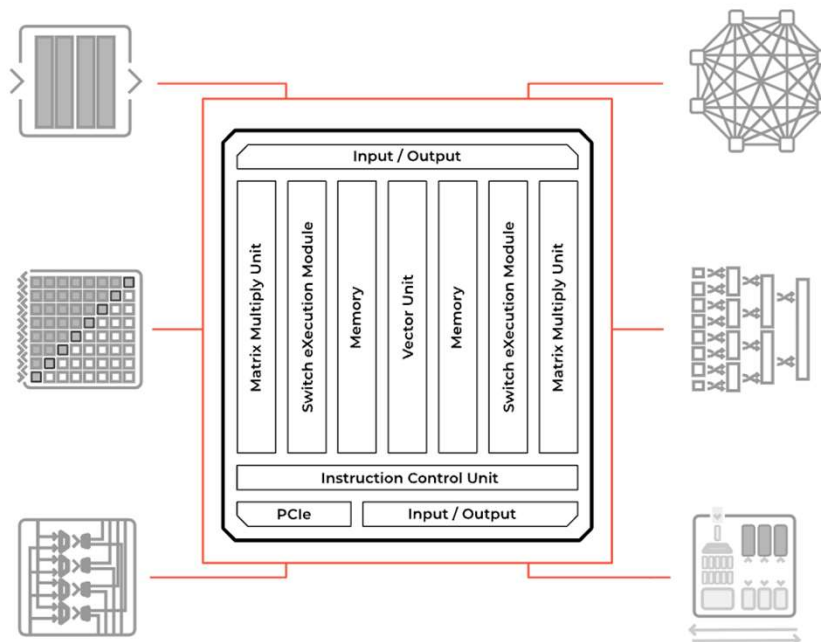
Massive concurrency
80 TB/s of BW
230MB capacity
Stride insensitive

Groq TruePoint™ Matrix

4x Engines
750 TOP/s int8
188 TFLOP/s fp16
320x320 fused dot product

Programmable Vector Units

5,120 Vector ALUs for high performance



Networking

480 GB/s bandwidth
Extensible network scalability
Multiple topologies

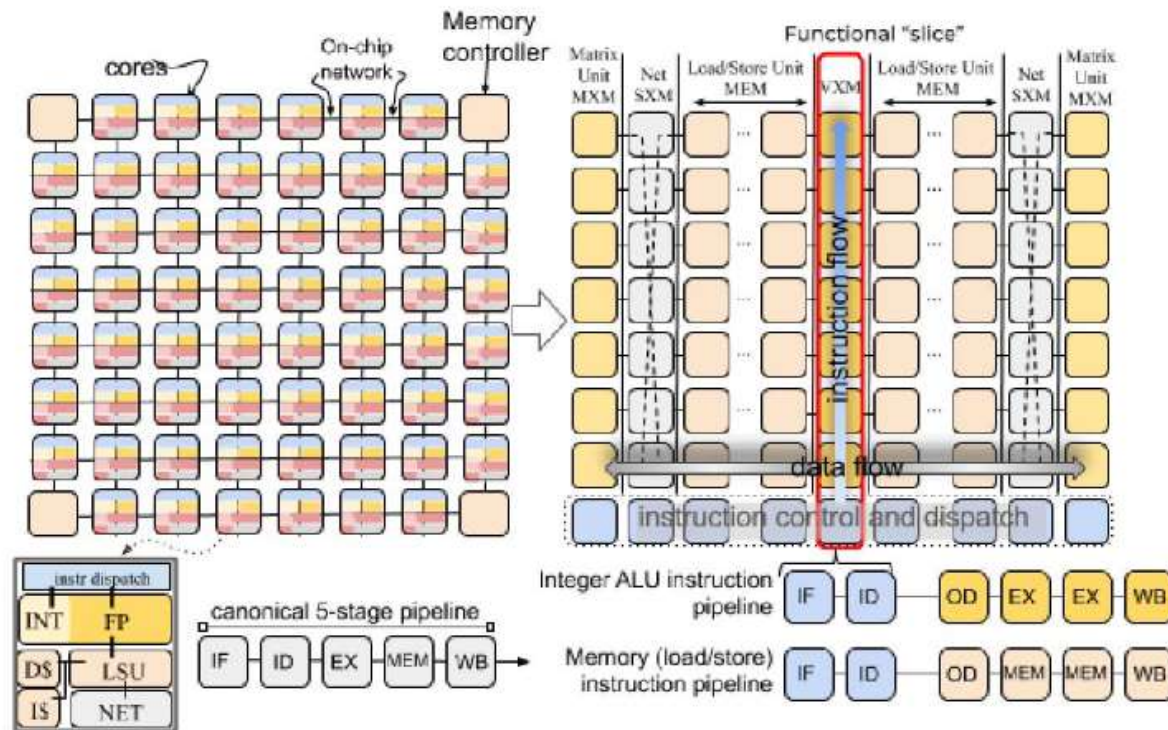
Data Switch

Shift, Transpose, Permuter for improved data movement and data reshapes

Instruction Control

Multiple instruction queues for instruction parallelism

Statički ili dinamički procesor (2)



TSP instrukcijski set [1]

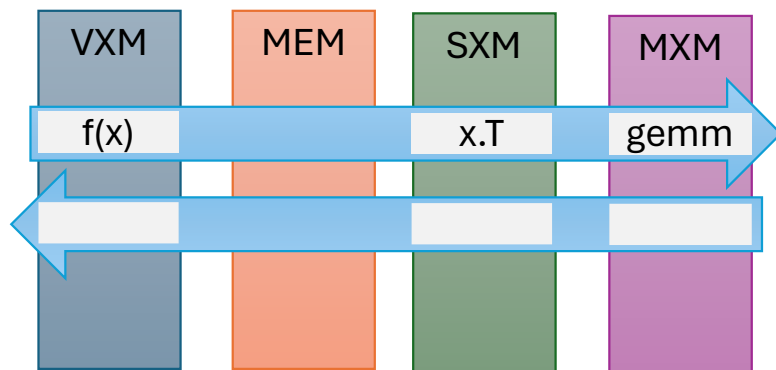
ICU	NOP N Ifetch Sync Notify Config Repeat n, d
MEM	Read a, s Write a, s Gather s, map Scatter s, map
VXM	unary operation binary operation type conversions ReLU TanH Exp RSqrt

MXM	LW IW ABC ACC
SXM	Shift <i>up/down</i> N Permute <i>map</i> Distribute <i>map</i> Rotate <i>stream</i> Transpose <i>sg16</i>
C2C	Deskew Send Receive

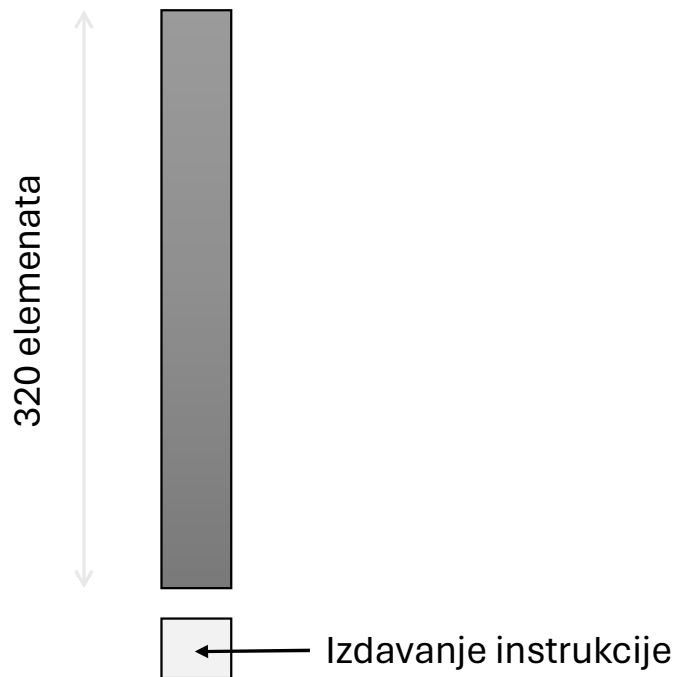
- Eksplicitna, kontrola na niskom nivou
- Matrice/vektori veličine 320 elemenata
- Eksplicitno selektovanje resursa

Tok podataka

- Podaci se premeštaju s jednog mesta na drugo
- Kompajler i programer organizuju to putovanje
- Posledica
 - Izračunavanje je posledica protoka podataka



Blokovi – SIMD jedinica



Različite SIMD jedinice

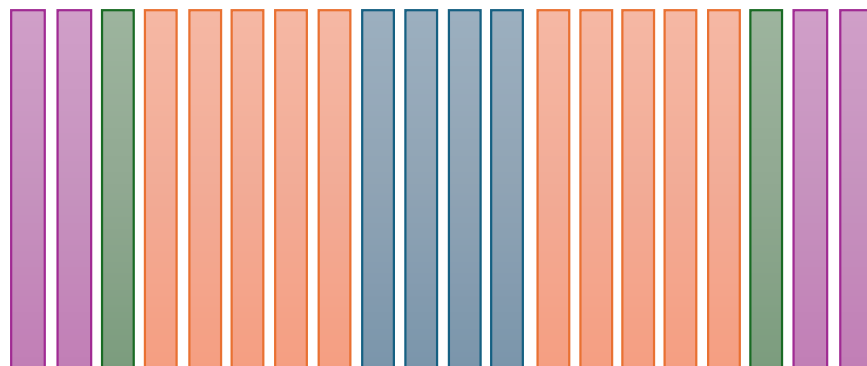
V
X
M

M
E
M

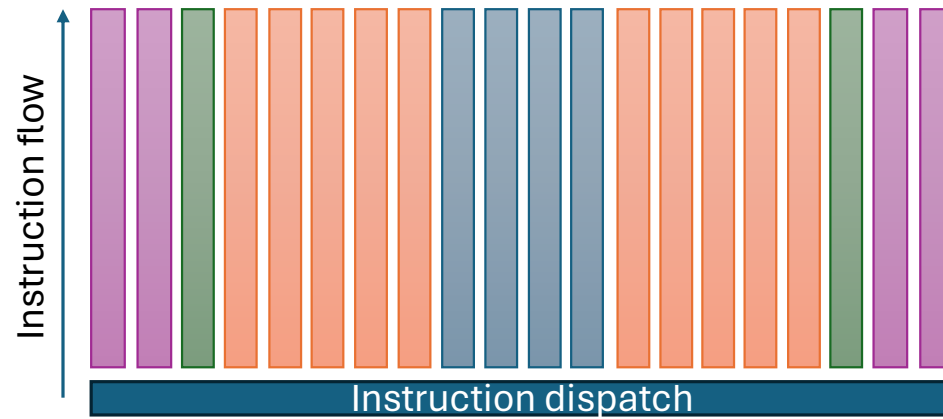
S
X
M

M
X
M

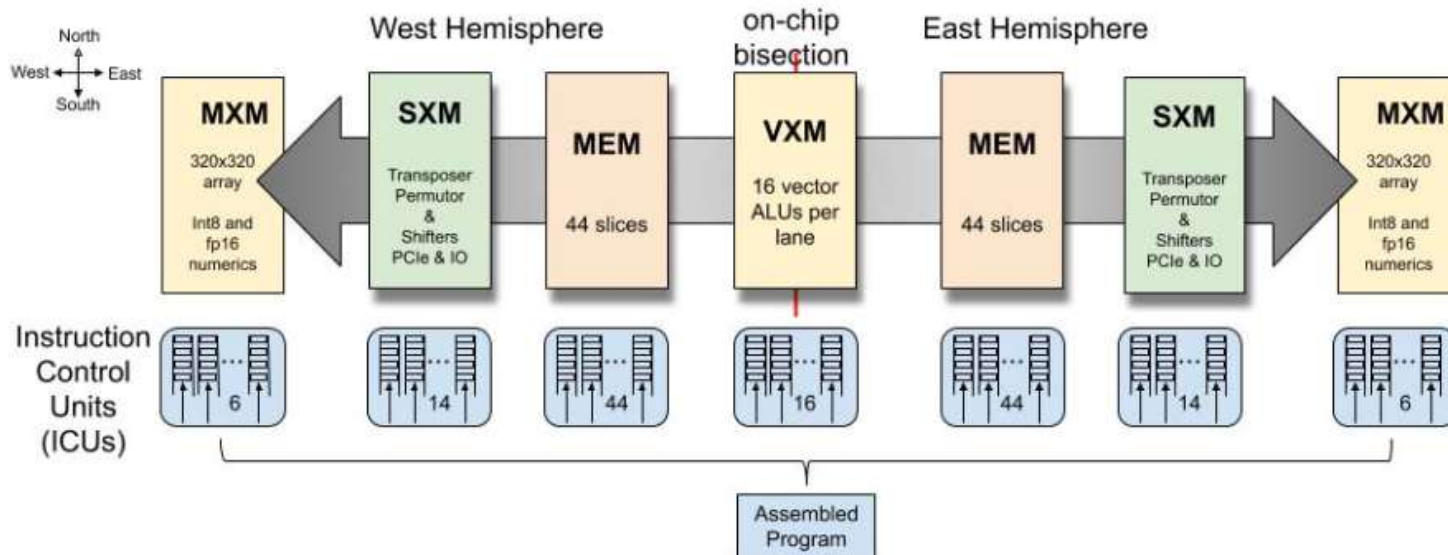
Duplicirati SIMD jedinice



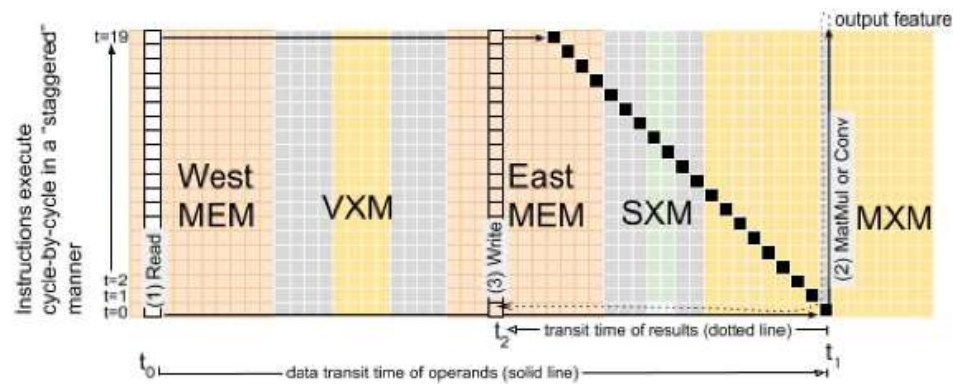
Instrukcije



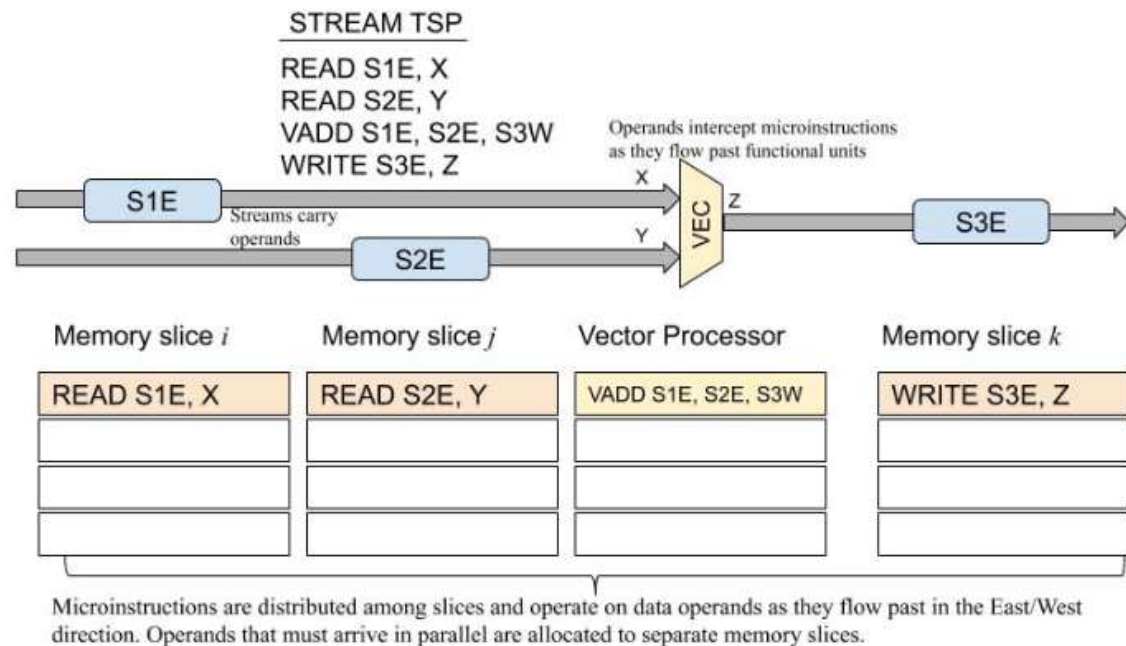
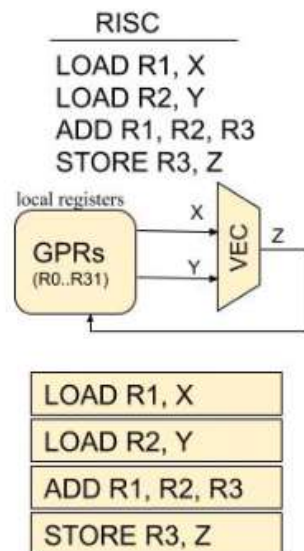
Izdavanje instrukcija



Izvršavanje instrukcija



Stream model izvršavanja (1)



Stream model izvršavanja (2)

- Strimovi su apstrakcija
- Strimovi su putanja kojom podaci teku kroz funkcionalne jedinice
- Proizvođač/potrošač model
- Kompajler precizno prati stanje čipa na nivou ciklusa
 - Tako obezbeđuje da instrukcija presretne operand u strimu
- Strimovi su implementirani pomoću registarskog fajla

Ispravljanje grešaka

- ECC sa 9 bitova
 - U strim registru se nalazi prostor za njih
 - Ispravlja se jedan bit
 - Dva bita se detektuju
- Provera se radi kod proizvođača u strimu
- Pokrivenost
 - Greške u memoriji
 - Greške u funkcionalnim jedinicama

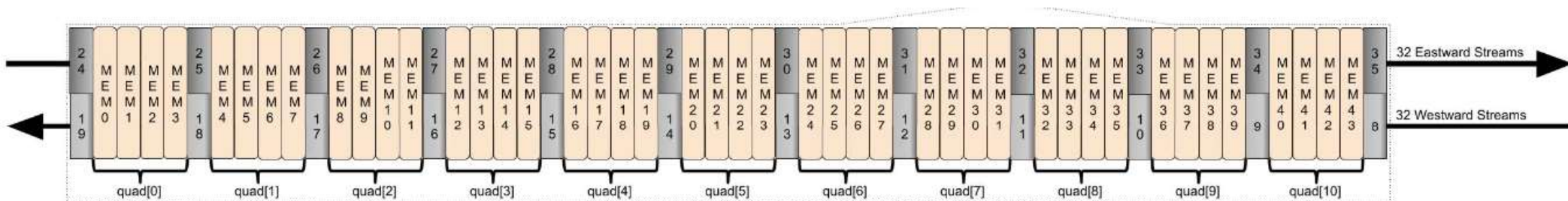
Organizacija



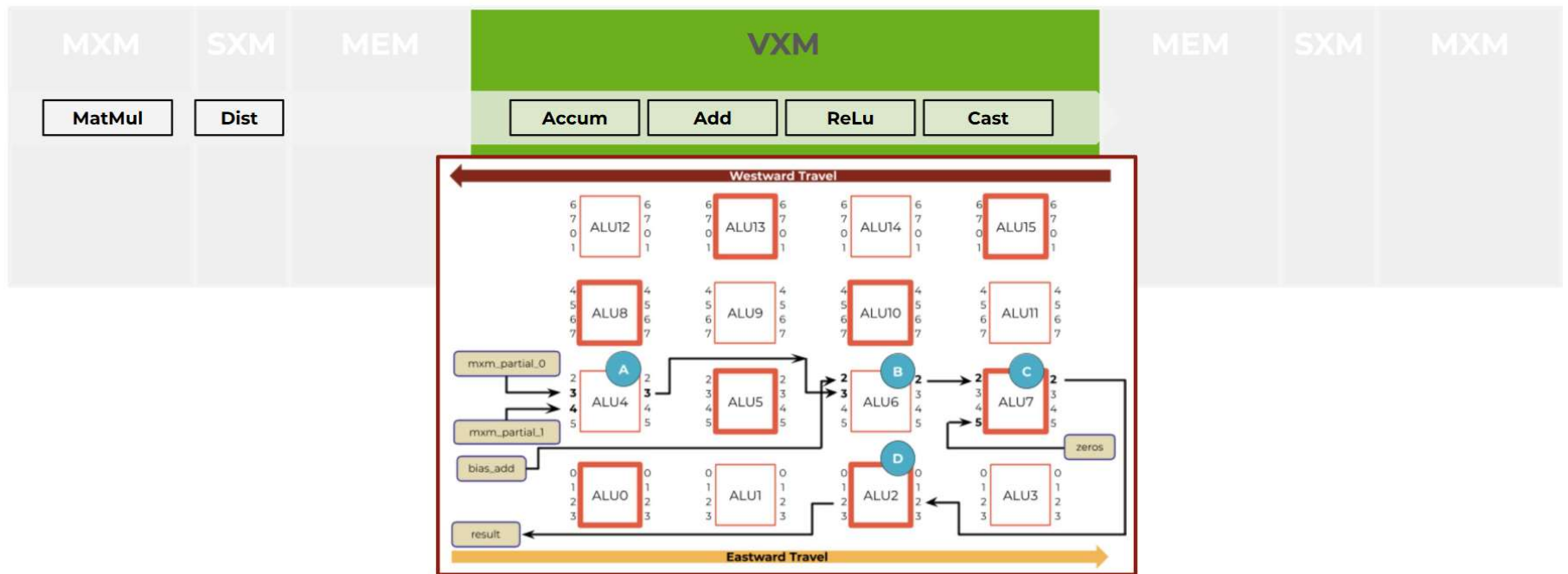
- 320 lejnova
 - Superlejn 16 lejnova
- 144 instrukcijska reda
- 64 logička strima po superlejnu
- SRAM 32 bajta po lejnu

Memorija

- Raspored u memoriji potpuno kontrolisan od strane softvera
- Kapacitet 220MB sa visokom propusnošću 55TBps
- Dovoljno za privremeno čuvanje podataka
- Nema DRAM-a, jer je nepredvidljiv

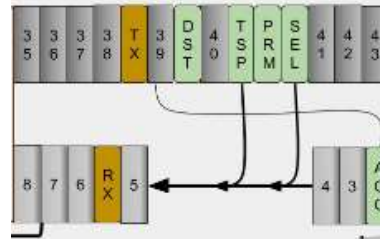


VXM

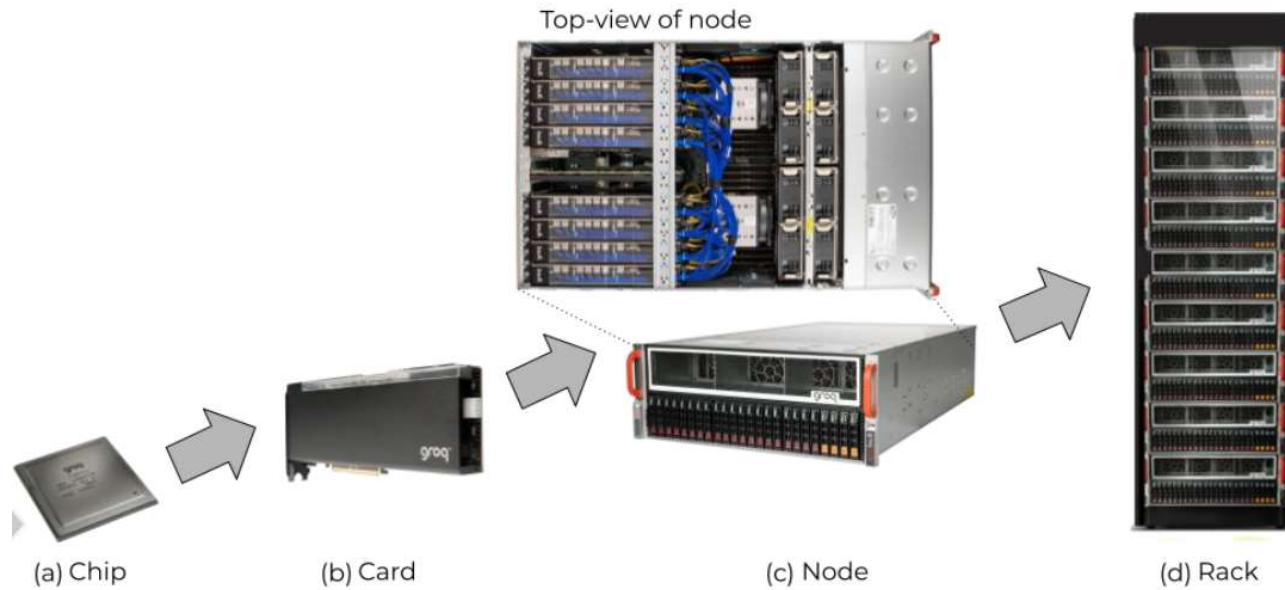


SXM

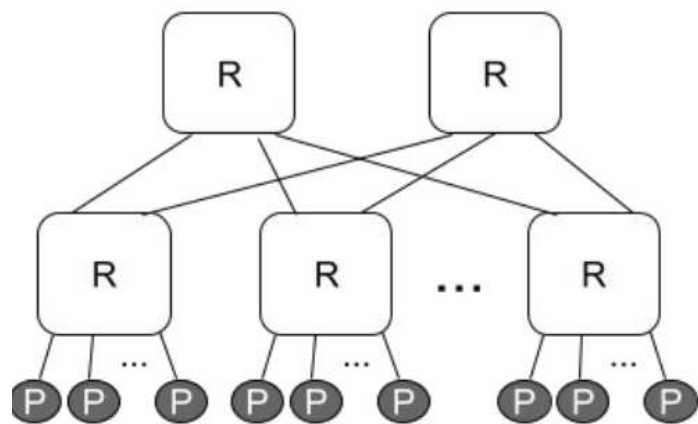
- Svajcarski nož za manipulisanjem podataka
 - Pomeranje, rotacija, distribuiranje, permutacije, transponovanje, distribucija
- IO
 - C2C
 - PCIe



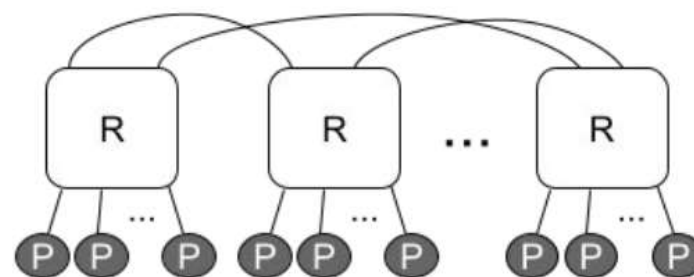
Skaliranje [2]



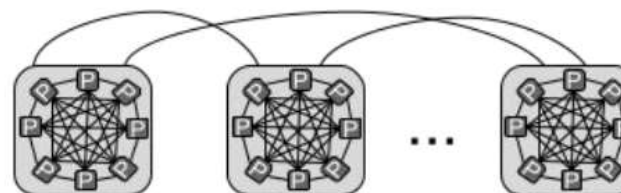
Interkonekcija



(a) Indirect network



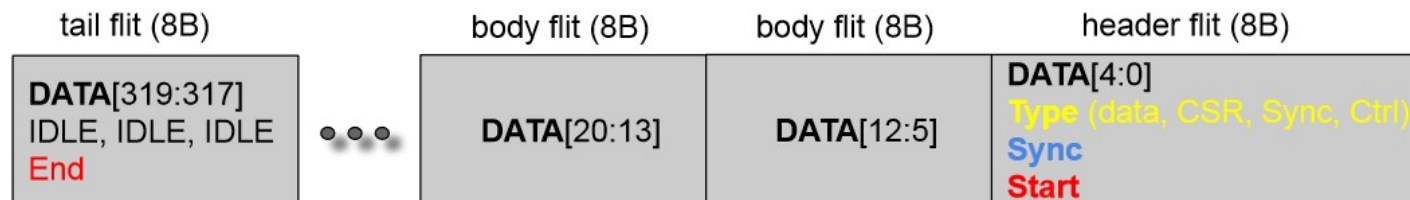
(b) Direct network



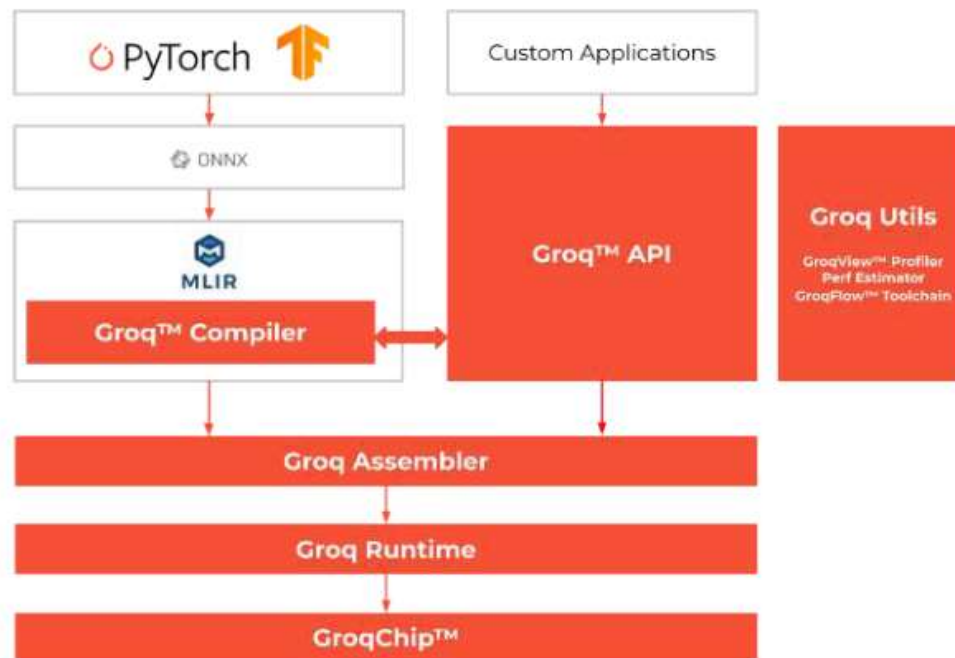
(c) Software-scheduled direct network

Determinizam u distribuiranom sistemu

- Težak problem
 - Klok ne dolazi iz istog izvora
- Hardverski handshake
 - Poravnanje brojača
- Nema rutiranja
 - Softver raspoređuje kako će se podaci kretati
- Paketi prilagođeni podacima koji se koriste na čipu



Softver



Primer koda

```
1 # Read from 16 slices onto 16 streams
2 # Transpose data
3 # Write from 16 streams into 16 slices
4 import groq as g
5 tensor = g.random_tensor(shape=[1024, 320],
6                             dtype=g.Int8, layout=[64, 16])
7 streams_16 = tensor.read(streams=range(16))
8 streams_16_t = g.transpose16(streams_16)
9 out_addrs = g.malloc(shape=[1024, 320],
10                        layout=[64, 16])
11 streams_16_t.write(out_addrs)
```

Reference

- [1] D. Abts et al., "Think Fast: A Tensor Streaming Processor (TSP) for Accelerating Deep Learning Workloads," ISCA, 2020
- [2] D. Abts et al., "A software-defined tensor streaming multiprocessor for large-scale machine learning, " ISCA, 2022
- [3] I. Ahmed, et al., "Answer Fast: Accelerating BERT on the Tensor Streaming Processor," ASAP, 2022